

Stored Cross Site Scripting (XSS) Attack Demonstration Using DVWA

Student Information

Experiment: Stored Cross Site Scripting (XSS)

Victim Server VM: cybersec_11

Attacker VM: cybersec_vm2

Vulnerable Application: DVWA (Damn Vulnerable Web Application)

Objective

To demonstrate a Stored Cross Site Scripting (XSS) attack by hosting DVWA on a victim server and accessing it from an attacker machine within a virtualized environment.

System Configuration

Victim Server

VM Name: cybersec_11

Web Server: Apache (XAMPP)

Database: MySQL (XAMPP)

Application: DVWA

IP Address: 192.168.70.130

Attacker Machine

VM Name: cybersec_vm2

IP Address: 192.168.70.131

Procedure

Step 1: Start the Victim Server

The victim server (cybersec_11) was started.

Apache and MySQL services were launched through XAMPP.

Screenshot

Step 2: Verify DVWA Availability

The DVWA application was accessed locally on the victim server.

URL:

<http://localhost/dvwa>

Screenshot

Step 3: Identify Victim Server IP Address

The IP address of the victim machine was identified using:

ipconfig

Victim IP:

192.168.70.130

Step 4: Access DVWA from Attacker Machine

The attacker machine (cybersec_vm2) accessed the victim server through:

http://192.168.70.130/dvwa

Successful access confirmed network connectivity between both virtual machines.

Screenshot

Step 5: Configure Security Level

The security level was configured to:

Low

This allows demonstration of the vulnerability.

Step 6: Launch Stored XSS Attack

The attacker navigated to:

XSS (Stored)

The following payload was entered:

alert('Stored XSS')

Step 7: Submit the Payload

The malicious JavaScript payload was submitted and stored by the application.

Screenshot

Step 8: Victim Accesses the Vulnerable Page

The victim opened the Stored XSS page after the attacker had injected the payload.

The malicious script executed automatically in the victim's browser.

Screenshot

Results

The Stored XSS attack was successfully executed.

Observations:

The attacker inserted malicious JavaScript into the application.

The application stored the payload.

The victim viewed the affected page.

The browser executed the stored JavaScript automatically.

The popup confirmed successful code execution.

Impact of the Vulnerability

Stored XSS vulnerabilities allow attackers to inject malicious scripts that execute in the browsers of other users.

Possible impacts include:

Session cookie theft

Account compromise

Defacement of web pages

User redirection to malicious websites

Execution of unauthorized actions

Conclusion

The Stored Cross Site Scripting (XSS) attack was successfully demonstrated using DVWA.

The attack showed that improperly validated user input can be stored by a web application and later executed in another user's browser. This highlights the importance of secure coding practices, input validation, output encoding, and regular security testing.